What Drives Chart-Based Return Prediction? Local Scaling, Temporal Learning, and Economic Limits

Abstract

We revisit what drives the strong return rankings produced by chart convolutional neural networks. Holding recent price and volume histories and portfolio rules fixed, we compare representations designed to assess the roles of local high–low scaling, binary image layout, and temporal learning. With a logistic classifier, local scaling makes a strong weekly ranking accessible before a flexible model is introduced. A one-dimensional convolutional neural network (CNN1D) applied to the continuous sequence then reproduces the chart CNN's weekly portfolio performance. On exact common stock-date panels, the model scores exhibit substantial but incomplete rank overlap; feature-level and prediction-level fusion add little to the strongest individual model. Learned portfolios remain strong under a common transaction-cost schedule, but net performance becomes negative under value weighting and modest tradability screens. Learned models and short-horizon price rules also weaken during 2020–2024. Most of the chart CNN's weekly portfolio performance can therefore be recovered from locally scaled continuous sequences with temporal convolution, without the binary image layout. Its measured economic value is concentrated in difficult-to-trade stocks and weakens after 2019.

Keywords: Return predictability, Machine learning, Representation learning, Market liquidity, Transaction costs

JEL classification: C45, G11, G12, G14

1. Introduction

Chart-based models raise a basic question for return prediction: when a machine learns from a market chart, what has it learned? A chart is not raw market history. Before a model sees it, the input window has been selected, prices have been scaled within that window, and price and volume information has been rendered in a fixed visual layout. A flexible model is then applied to the resulting representation. The strong portfolio performance of chart convolutional neural networks (chart CNNs) (Jiang et al. 2023) may therefore reflect the chart image, the local high–low scaling embedded in chart construction, or nonlinear learning from recent price-volume information.

Recent market history can be represented through prespecified price rules, ordered numerical sequences, or chart images. Lo et al. (2000) show how chart patterns can be defined and tested systematically, while Murray et al. (2024) study nonlinear predictability in numerical return histories. Evidence that carefully constructed linear forecasts can match more complex machine-learning methods further underscores the need to distinguish predictive inputs from model complexity (Cheng et al. 2025). Jiang et al. (2023) provide the direct starting point for our analysis. Beyond applying a CNN to price and volume images, they compare logistic classifiers and CNN1D across image, cumulative-return, and devolatized-return scaling for all nine input-window and forecast-horizon combinations under equal and value weighting. Jiang et al. already show that local scaling and CNN1D can reproduce much of the chart result at the portfolio level. We take that comparison as our starting point and ask whether the models rank the same stocks, whether their representations contain complementary information, and where the resulting signal remains economically implementable.

To examine what drives chart-CNN performance, we compare logistic, CNN1D, and chart-CNN rankings through 2024 while holding recent market histories and portfolio rules fixed. We then match the learned-model scores stock by stock and date by date. On these common panels, we calculate rank correlations and estimate joint encompassing regressions. We also combine the chart and sequence inputs at the feature and prediction levels to see whether they improve one another. Model development remains confined to 1993–2000, and all portfolio results are evaluated out of sample over 2001–2024 under common ranking, weighting, holding-period, and transaction-cost rules.

Most of the chart CNN's weekly portfolio performance can be recovered from locally scaled continuous sequences with temporal convolution. In the representative $I20/R5$ specification, the image-scale logistic classifier generates an equal-weighted gross Sharpe ratio of 5.22, compared with -0.41 under cumulative-return scaling. CNN1D raises the Sharpe ratio to 6.55, close to 6.52 for the chart CNN, and remains competitive across the other weekly input windows. Although their portfolio Sharpe ratios are nearly identical, the models rank stocks similarly but not identically. Neither feature-level nor prediction-level fusion consistently improves on the strongest individual representation. The binary chart can therefore retain score-level variation without adding material portfolio performance once the same local scaling and temporal structure are available.

We also reassess Jiang et al.'s economic conclusions using the same four price-based benchmarks and a common stylized transaction-cost schedule. Related cross-sectional evidence finds strong decile-portfolio performance from random-forest rankings built on technical indicators in a liquid international stock universe (Breitung 2023). Full-universe equal-weighted portfolios retain net Sharpe ratios near four, consistent with Jiang et al.'s finding that stated costs do not eliminate portfolio performance. Net returns turn negative under our dollar-volume and market-capitalization screens and under value weighting. The chart CNN, CNN1D, multimodal model, and short-horizon price rules also weaken during 2020–2024. Historical performance is thus concentrated in equal-weighted portfolios and stocks that are difficult to trade at scale, and its magnitude attenuates after 2019. This pattern is consistent with recent evidence on implementation constraints and decay in chart-CNN performance (Kaczmarek and Pukthuanthong 2025).

Our contribution is to connect Jiang et al.'s representation comparison to stock-level and implementation evidence. We test whether similar portfolio outcomes arise from the same rankings, whether image and sequence inputs contain complementary information, and whether the common signal survives basic economic boundaries. This shifts the question from which architecture performs best to which components are sufficient to recover the observed portfolio performance. Section 2 presents the controlled design. Section 3 examines the mechanism and score-level results. Section 4 assesses benchmark performance, implementation, and post-2019 stability. Section 5 concludes.

2. Data, inputs, and empirical design

The prediction pipeline combines local high–low scaling and a binary chart representation with flexible learning. Building on the controlled representation comparison in Jiang et al. (2023), we compare logistic, CNN1D, and chart-CNN rankings under alternative scaling schemes. We then place the learned-model scores on exact common stock-date panels and measure ranking overlap, conditional predictive content, and multimodal complementarity. Timing, ranking, weighting, and transaction-cost rules remain fixed across the learned-model evaluations.

2.1. *Sample and prediction target*

We construct a daily CRSP panel of ordinary common shares listed on NYSE, AMEX, or Nasdaq using share codes 10 and 11. The panel includes daily returns, open, high, low, and close prices, trading volume, lagged market capitalization, stock identifiers, and available delisting returns. Eligibility requires positive prices and sufficient history for the relevant specification. Firms enter and leave with their CRSP histories, producing an unbalanced panel without a survivorship requirement.

Model development uses data from 1993–2000, and portfolio evaluation is out of sample from 2001 through 2024. We report the full evaluation period, a 2001–2019 baseline sample, and a 2020–2024 extension sample. The model grid combines input windows $I \in \{5, 20, 60\}$ with future return horizons $R \in \{5, 20, 60\}$. We write a 20-day input and 5-day target as $I20/R5$. For stock i at formation date t, the directional target is

$$y_{i,t}^{(R)} = \mathbb{1}\left(\prod_{j=1}^{R}(1 + r_{i,t+j}) - 1 > 0\right). \tag{1}$$

The target begins at $t + 1$, after all input information has been observed by the close of date t.

To remove mechanical price jumps from stock splits, we set the first close in each input window to one, reconstruct subsequent closes from daily returns, and scale open, high, and low prices consistently with the reconstructed close series. The moving-average window equals I. Within a specification, the learned models use the same underlying observations. Eligible cross-sections can differ across input windows and price-based benchmarks because their history requirements differ. Portfolio results use the sample available to each score, while direct learned-model overlap tests impose an exact common stock-date panel.

2.2. *Input construction and representations*

The chart input combines local high–low scaling with a binary grayscale representation. We refer to the underlying within-window transformation as *local high–low scaling*; tables retain Jiang et al.'s label, *image scale*, for the corresponding continuous sequence input. For each stock and formation date, let $m_{i,t}^{(I)}$ and $M_{i,t}^{(I)}$ denote the minimum and maximum calculated jointly across all open, high, low, close, and moving-average observations in the lookback window. Each price-channel observation is transformed as

$$\widetilde{x}_{i,t-j} = \frac{x_{i,t-j} - m_{i,t}^{(I)}}{M_{i,t}^{(I)} - m_{i,t}^{(I)}}. \tag{2}$$

This transformation removes the nominal price level and normalizes the magnitude of the window's price range while preserving each observation's relative position and the path through that range. Trading volume is scaled separately within the same window. Chart construction then places daily OHLC bars and a moving-average line in the price panel and volume in a lower panel. The 5-, 20-, and 60-day images contain 32×15, 64×60, and 96×180 binary pixels, respectively. The online appendix provides an input illustration and the complete representation map.

The numerical sequence contains the same six daily channels in chronological order: open, high, low, close, moving average, and volume. We evaluate three sequence specifications. The image scale retains the continuous within-window values from Eq. (2); the cumulative-return scale anchors each price channel to the first close; and the devolatized-return scale expresses price changes relative to lagged exponentially weighted volatility and volume through relative changes. Each transformed channel is standardized using development-sample moments that remain fixed during evaluation.

The chart and locally scaled sequence therefore share the same local high–low scaling but differ in representation. The sequence retains continuous observations ordered by trading day, whereas the chart converts them into a fixed binary pixel layout. We use logistic classifiers to measure how each input transformation changes a linear ranking. We then compare the logistic classifier with CNN1D under image scaling to measure what nonlinear temporal learning adds, and CNN1D with the chart CNN to determine whether similar portfolio performance requires the binary image.

The chart CNN and CNN1D differ not only in image layout but also in numerical precision, dimensionality, filter geometry, depth, and capacity. We therefore cannot interpret their comparison as a pure treatment of image layout. Similar performance after local high–low scaling means that the portfolio pattern can be recovered without the binary chart image, while any residual difference may come from the other representation or architecture features.

2.3. Prediction models and estimation

The chart CNN follows the design of Jiang et al. (2023) and applies two-dimensional convolutions to the binary images. CNN1D applies one-dimensional convolutions along the trading-day dimension of the six-channel sequence. Both architectures increase in depth with the input window. The logistic specifications flatten the sequence into a linear index. The online appendix reports the complete architecture, scaling, and training settings.

The multimodal model pairs chart and sequence inputs for the same stock, formation date, input window, and forecast horizon. Separate chart CNN and CNN1D branches produce feature vectors that are concatenated before a linear classification head. We also examine late fusion by averaging the predictions of separately trained branches. Each model's portfolio score is the softmax probability assigned to the positive-return class; it is used for within-date ranking and is not interpreted as a calibrated expected return.

We estimate separate models across input windows, forecast horizons, model families, sequence scales, and fusion designs. Neural models minimize cross-entropy loss with Adam and stop when development-sample validation loss fails to improve for two consecutive epochs. Stock-date observations from 1993–2000 are randomly divided into 70% training and 30% validation sets using a fixed seed, and the lowest-validation-loss checkpoint remains fixed throughout the 2001–2024 evaluation period. The main analysis uses one selected checkpoint per specification. The appendix reports all estimation settings, a five-member chart-CNN sensitivity check for the $I5/R5$ and $I20/R5$ specifications, and a compact comparison of verified design similarities and differences relative to Jiang et al.

2.4. Portfolio construction and empirical tests

At each formation date, the eligible cross-section is sorted into ten deciles using the relevant model score or benchmark signal. The high-minus-low portfolio buys the highest-score decile and shorts the lowest-score decile. Equal-weighted portfolios assign the same weight to every stock within a leg; value-weighted portfolios use lagged market capitalization normalized separately within each leg. Each leg carries one dollar of notional, so the spread has zero net investment and two dollars of gross exposure.

Weekly, monthly, and quarterly portfolios correspond to the 5-, 20-, and 60-trading-day forecast horizons. Information through the close of date t ranks stocks for the return from $t+1$ through $t+R$. One formation date per holding interval produces non-overlapping return series. For a portfolio return series $R_{p,t}$ with $K \in \{52, 12, 4\}$ observations per year, the annualized Sharpe ratio is

$$\mathrm{SR}_p = \sqrt{K}\,\frac{\overline{R}_p}{\mathrm{sd}(R_{p,t})}. \tag{3}$$

No risk-free rate is subtracted from the zero-net-investment spread.

We report the full grid across input windows and forecast horizons. The later economic tests use $I20/R5$ as the representative weekly specification because it is central to the short-horizon grid and performs strongly across all three learned model families. This specification was selected after examining the grid, so the focused comparisons are descriptive. Benchmark signals can require longer histories than the learned models, and standalone portfolios therefore need not contain identical stock-date observations.

Price-based benchmarks. We compare the learned rankings with momentum (MOM), one-month short-term reversal (STR), one-week short-term reversal (WSTR), and the multi-horizon trend strategy (TREND). MOM is a daily analogue of 12–1 momentum (Jegadeesh and Titman 1993); STR and WSTR use the negative of the previous 21- and 5-trading-day returns (Jegadeesh 1990; Lehmann 1990); and TREND combines moving-average-to-price ratios across horizons using lagged forecasting coefficients (Han et al. 2016). All benchmarks follow the common portfolio protocol. The appendix reports their complete formulas and result grids.

Costs and implementation. Net performance applies one-way costs to full absolute traded notional in both legs: 10 basis points for stocks above the contemporaneous NYSE 80th-percentile market-capitalization breakpoint and 20 basis points otherwise. Reported turnover is conventional half-turnover summed across the two legs and expressed as a monthly equivalent. The cost schedule excludes stock-specific spreads, market impact, borrowing costs, short-sale constraints, and capacity limits. We also re-form the representative portfolios after price, dollar-volume, and NYSE market-capitalization screens while leaving model scores fixed. Carhart four-factor regressions use weekly net high-minus-low returns, annualized intercepts, and Newey–West standard errors with four lags (Carhart 1997).

Ranking overlap and conditional content. For each input window and forecast horizon, we form an exact common panel containing all three learned-model scores and the future return. Date-level Spearman correlations measure similarity in the within-date rankings (Spearman 1904). We also estimate the cross-sectional regression

$$R_{i,t+1:t+R} = \alpha_t + \beta_t^{2D} s_{i,t}^{2D} + \beta_t^{1D} s_{i,t}^{1D} + \beta_t^{MM} s_{i,t}^{MM} + \varepsilon_{i,t+R}. \tag{4}$$

The Fama–MacBeth coefficients average the date-level slopes through time (Fama and MacBeth 1973). Their conventional test statistics use the time-series standard errors of the slopes without an autocorrelation correction and are treated as exploratory, especially at longer horizons. Because score scales differ across separately trained models, we emphasize coefficient signs and patterns that recur across specifications.

The empirical grid contains related comparisons across input windows, forecast horizons, scaling schemes, model architectures, fusion designs, weighting rules, and sample periods. We do not apply a formal multiple-testing correction. We therefore emphasize patterns that recur across specifications and across the benchmark, transaction-cost, tradability, and extension-sample results rather than individual cells.

3. What drives chart-CNN performance?

Jiang et al. (2023) compare logistic classifiers and CNN1D across alternative scaling schemes and show that image-scaled CNN1D can match the portfolio performance of the chart CNN. We begin with the same scaling comparison through 2024. We then compare the model scores stock by stock and combine chart and sequence inputs to determine whether the two representations rank different firms or improve one another.

3.1. Local scaling and temporal learning

Table 1 re-estimates Jiang et al.'s specifications over the extended 2001–2024 evaluation sample. Panel A compares the logistic classifier across cumulative-return, devolatized-return, and image scaling. Panel B reports image-scaled CNN1D and chart-CNN results alongside the multimodal extension. Complete monthly and quarterly results appear in the online appendix.

Table 1: Extended-sample re-estimation of the weekly representation comparison

	5-day input		20-day input		60-day input	
Model	EW	VW	EW	VW	EW	VW
Panel A. Encoding with a logistic classifier						
Logistic, cumulative scale	0.18	−0.04	−0.41	−0.51	−0.26	−0.53
Logistic, devolatized scale	2.80	0.79	2.73	0.48	2.70	0.66
Logistic, image scale	4.88	1.25	5.22	1.09	5.44	1.12
Panel B. Flexible models with image scale						
CNN1D	6.24	1.16	6.55	1.12	5.89	2.06
Chart CNN	5.88	1.39	6.52	1.44	3.72	1.18
Multimodal	6.01	1.16	6.48	1.52	5.53	1.75

Note. The table reports annualized gross Sharpe ratios of non-overlapping 5-day high-minus-low decile portfolios from 2001 through 2024. Panel A and the CNN1D and chart-CNN rows in Panel B re-estimate the corresponding specifications in Jiang et al. (2023), Table IX, over the extended evaluation sample; the multimodal row is an extension. EW and VW denote equal- and value-weighted legs. Image scale is the specification label for local high–low scaling of the price and moving-average channels; volume is scaled separately. Chart CNN receives the corresponding binary image. Multimodal is the feature-concatenation model. Each row uses that model's available score cross-section; the rows are not restricted to the exact three-model stock-date intersection used in the overlap tests. The online appendix reports the full scaling and return-horizon grids.

In the representative $I20/R5$ specification, the logistic Sharpe ratio increases from −0.41 under the cumulative-return scale to 5.22 under the image scale, with the devolatized-return scale between them. The same ordering holds across the other input intervals. By preserving the relative position of prices within each recent high–low range, local high–low scaling makes a strong return ranking accessible even to a linear classifier.

Temporal convolution applied to the locally scaled sequence closely matches the chart CNN's weekly portfolio performance. In the representative $I20/R5$ specification, the Sharpe ratio increases from 5.22 for the image-scale logistic classifier to 6.55 for CNN1D, which is nearly identical to 6.52 for the chart CNN. CNN1D remains competitive across the other input intervals. Thus, temporal learning can recover the chart CNN's weekly portfolio performance without the binary pixel layout.

At longer horizons, local high–low scaling remains the strongest logistic representation, but temporal convolution adds little. In $I20/R20$, the image-scale logistic classifier reaches a Sharpe ratio of 2.37, compared with 2.15 for CNN1D, and the difference becomes more pronounced as quarterly portfolio performance weakens. Local high–low scaling therefore remains informative across horizons, while temporal learning contributes most at the weekly horizon.

Value weighting sharply reduces the weekly result. In $I20/R5$, Sharpe ratios near 6.5 for the three equal-weighted learned portfolios fall to between 1.12 and 1.52 under value weighting. The models remain close to one another, but the smaller economic magnitude indicates that much of the weekly spread comes from stocks receiving little weight in value-weighted portfolios. We examine this concentration directly with tradability screens in Section 4.2.

In the representative $I20/R5$ profiles, all three learned models produce broadly similar and mostly increasing decile returns. Equal-weighted annualized returns range from approximately -35% to -32% in the lowest decile and from 53% to 55% in the highest. The especially steep increase from decile 9 to decile 10 places additional return in the highest-ranked tail. Value-weighted profiles are flatter and receive little contribution from the lowest-score portfolio, again pointing to a strong role for smaller stocks.

CNN1D applied to the locally scaled sequence reproduces the chart CNN's weekly performance, but differences in input precision, filter geometry, and depth can still alter individual scores. Similar Sharpe ratios therefore need not imply that the models rank the same stocks. We compare their scores directly in the next subsection.

3.2. Representation overlap and conditional content

Jiang et al.'s Table IX compares portfolio performance but does not report direct rank correlations between chart-CNN and CNN1D scores. We calculate average date-level Spearman correlations on the exact common stock-date panel and report them in Table 2. The correlations measure how similarly the models order stocks within each formation date.

Rank overlap is strongest in the weekly specifications where portfolio performance is strongest. For $I5/R5$ and $I20/R5$, pairwise Spearman correlations range from 0.67 to 0.84. The similar weekly portfolio results are therefore accompanied by substantial overlap in the underlying stock rankings. The correlations summarize the full cross-section, whereas the long–short portfolio uses only the highest and lowest deciles. Models can consequently produce similar Sharpe ratios if they form similar tail portfolios even while ordering many middle-ranked stocks differently.

The models diverge most clearly with the 60-day input. In $I60/R5$, the chart CNN/CNN1D rank correlation falls to 0.44 as their Sharpe ratios separate to 3.72 and 5.89. CNN1D also preserves its weekly ranking more consistently across input windows than the chart CNN. The representation difference is therefore more visible with the longer lookback window than with the 5- and 20-day inputs.

Rank correlations do not reveal whether one model's score remains related to future returns after conditioning on the other scores. We estimate that relationship with joint cross-sectional regressions and report the Fama–MacBeth t-statistics in Table 3.

In $I5/R5$ and $I20/R5$, all three scores remain positively related to future returns after conditioning on one another. Their overlapping rankings therefore still contain model-specific predictive

Table 2: Rank overlap across learned models and input windows

Specification	Chart/CNN1D	Chart/Multimodal	CNN1D/Multimodal
Panel A. Across-model rank overlap			
$I5/R5$	0.73	0.74	0.84
$I20/R5$	0.67	0.72	0.80
$I60/R5$	0.44	0.65	0.69
$I5/R20$	0.50	0.67	0.71
$I20/R20$	0.53	0.65	0.72
$I60/R20$	0.44	0.55	0.63
$I5/R60$	0.45	0.45	0.68
$I20/R60$	0.44	0.45	0.64
$I60/R60$	0.38	0.46	0.63
Panel B. Within-model rank stability for the 5-day return horizon			
Model	$I5/I20$	$I5/I60$	$I20/I60$
Chart CNN	0.60	0.22	0.34
CNN1D	0.64	0.54	0.71
Multimodal	0.56	0.44	0.59

Note. Panel A reports time-series averages of formation-date Spearman correlations on the common stock-date panel for which all three learned scores and the future return are observed. Panel B compares scores from two input windows within the same model family. All statistics use the 2001–2024 evaluation window.

Table 3: T-statistics from joint encompassing regressions

Specification	Chart CNN	CNN1D	Multimodal	Average R^2 (%)
$I5/R5$	6.95	12.56	8.39	0.70
$I20/R5$	9.51	11.80	11.91	0.80
$I60/R5$	−1.12	18.19	9.95	0.95
$I5/R20$	−0.26	4.39	−0.73	0.52
$I20/R20$	3.93	5.76	3.36	0.65
$I60/R20$	−0.85	4.66	0.28	1.06
$I5/R60$	−0.74	2.16	2.56	0.50
$I20/R60$	−0.50	2.83	1.37	0.75
$I60/R60$	−0.46	3.97	0.71	1.13

Note. The first three columns report the conventional Fama–MacBeth t-statistic for each score in a joint date-level cross-sectional regression over 2001–2024. The statistic uses the time-series standard error $\mathrm{sd}(\beta_t)/\sqrt{T}$ without an autocorrelation correction. It is therefore exploratory, especially at longer horizons and in view of the model search. The last column is the time-series average of the date-level joint-regression R^2.

variation. In $I60/R5$, the chart statistic becomes weak while CNN1D remains large, consistent with the earlier separation in rankings and portfolio performance. CNN1D is positive throughout the full horizon grid and is the most consistent conditional predictor.

3.3. Complementarity across representations

Combining chart and sequence information does not improve weekly portfolio performance. In the central $I20/R5$ specification, the multimodal model reaches a Sharpe ratio of 6.48, compared with 6.52 for the chart CNN and 6.55 for CNN1D, and the best single branch remains strongest in the other weekly specifications. The same pattern appears after separate training: averaging the chart CNN and CNN1D probabilities or logits produces weekly performance close to the individual branches. Detailed comparisons appear in the online appendix.

Both fusion designs preserve strong weekly portfolio performance but fail to improve on the best individual branch. Chart and sequence representations therefore overlap substantially but incompletely and offer little portfolio-level complementarity.

4. Economic value: implementation and fragility

The learned models rank stocks strongly in the unrestricted equal-weighted universe. We now ask how much of that performance survives transparent benchmarks, trading costs, tradability screens, and the 2020–2024 period. A spread that disappears under value weighting or modest liquidity restrictions has a different economic meaning from one that remains available among larger, more tradable stocks.

4.1. Gross performance and the cost hurdle

Table 4 compares the representative $I20/R5$ learned portfolios with MOM, STR, WSTR, and TREND over 2001–2024. Each method retains its own score definition and lookback, but all enter the same weekly non-overlapping decile construction, equal-weighting rule, and transaction-cost adjustment. The rows are therefore comparable as portfolio strategies, although method-specific histories can change the eligible stock-date samples.

Table 4: Weekly equal-weighted learned models and price-based benchmarks

Method	Gross return	Gross SR	Net return	Net SR	Turnover
Chart CNN $I20/R5$	84.9%	6.52	51.3%	3.94	6.64
CNN1D $I20/R5$	89.1%	6.55	56.5%	4.16	6.46
Multimodal $I20/R5$	87.5%	6.48	54.5%	4.04	6.56
MOM	−2.3%	−0.08	−7.1%	−0.25	0.95
STR	44.3%	1.83	27.3%	1.13	3.32
WSTR	59.3%	2.86	24.9%	1.20	6.71
TREND	52.7%	2.49	27.5%	1.31	4.95

Note. Returns and Sharpe ratios are annualized. Turnover is monthly-equivalent conventional half-turnover summed across the two legs. The transaction-cost deduction uses full absolute traded notional. Learned and benchmark strategies draw from the same CRSP source universe, though method-specific lookback requirements can change eligible stock counts.

The learned portfolios clear the gross benchmark hurdle by a wide margin: their Sharpe ratios are near 6.5, compared with 2.86 for WSTR, the strongest transparent rule. Their turnover is also close to WSTR's. Under the common 10–20 basis-point schedule, trading reduces learned-model annualized returns by roughly 33 percentage points, but their larger gross spreads still leave net Sharpe ratios near four; TREND is the strongest net benchmark at 1.31.

These numbers apply only to equal-weighted portfolios under a stylized cost schedule. Under value weighting, all three learned portfolios retain positive gross Sharpe ratios, but their net Sharpe ratios are negative in both the baseline and extension samples. The cost schedule also excludes market impact, borrowing costs, short-sale constraints, and capacity. The learned models therefore rank stocks strongly in the unrestricted universe, but the estimates do not establish scalable profitability. Their advantage also narrows at longer horizons and when larger stocks receive more weight.

4.2. *Performance under tradability screens*

We keep the $I20/R5$ scores fixed and re-form the portfolios after price, dollar-volume, and market-capitalization screens. Each filter is applied before sorting. Table 5 therefore shows whether the same ranking remains economically useful among more tradable stocks.

Table 5: Tradability and factor diagnostics for representative weekly portfolios

Model	Screen	Net return	Net SR	Carhart α	$t(\alpha)$	Names	Turnover
Chart CNN	Full universe	51.3%	3.94	49.3%	11.81	420	6.65
	Price ≥ \$5	10.1%	1.08	8.1%	3.67	321	6.66
	Dollar volume ≥ \$1m	−10.0%	−0.85	−11.0%	−3.87	241	7.01
	Dollar volume ≥ \$5m	−19.5%	−1.57	−20.6%	−7.27	164	7.07
	No NYSE microcaps	−17.0%	−1.60	−18.1%	−7.38	193	6.89
	Combined	−22.9%	−2.10	−24.0%	−10.21	149	7.02
CNN1D	Full universe	56.5%	4.16	55.2%	12.56	419	6.46
	Price ≥ \$5	13.7%	1.48	12.4%	5.46	319	6.50
	Dollar volume ≥ \$1m	−5.7%	−0.46	−6.2%	−2.21	240	6.68
	Dollar volume ≥ \$5m	−15.0%	−1.17	−15.7%	−5.61	164	6.75
	No NYSE microcaps	−14.8%	−1.37	−15.6%	−6.34	192	6.57
	Combined	−19.0%	−1.68	−19.8%	−8.11	149	6.69
Multimodal	Full universe	54.5%	4.04	53.2%	12.10	419	6.56
	Price ≥ \$5	13.5%	1.52	12.4%	5.46	319	6.63
	Dollar volume ≥ \$1m	−5.2%	−0.42	−5.5%	−1.91	240	6.83
	Dollar volume ≥ \$5m	−14.1%	−1.11	−14.7%	−5.12	164	6.92
	No NYSE microcaps	−14.0%	−1.33	−14.6%	−5.85	192	6.75
	Combined	−19.3%	−1.76	−19.9%	−8.32	149	6.87

Note. The table reports equal-weighted $I20/R5$ portfolios from 2001 through 2024. Names is the average number of stocks in each leg. Combined requires price of at least \$5, dollar volume of at least \$5 million, and market capitalization above the NYSE 20th-percentile breakpoint. Carhart intercepts are annualized and use Newey–West standard errors with four lags.

Net performance falls quickly as liquidity requirements tighten. A \$5 price screen leaves positive but much smaller returns, while a \$1 million dollar-volume screen makes all three portfolios negative. Excluding NYSE microcaps or combining the screens also produces negative returns, and turnover remains high. CNN1D, for example, moves from a 56.5% net annualized return in the full universe to −5.7% at the \$1 million dollar-volume threshold and −19.0% under the combined screen.

Jiang et al. (2023) report positive results among the largest 500 stocks, whereas our portfolios become negative under alternative tradability criteria. The exercises are not directly equivalent: Jiang et al. rank stocks within a largest-500 universe, while our filters use price, dollar-volume, and NYSE market-capitalization cutoffs; the evaluation period and model-estimation conventions also differ. Online Appendix Table A.2 lists these design differences. Our screens therefore qualify, rather than directly overturn, their implementation conclusion.

Carhart adjustment leaves the unrestricted spread largely intact: the full-universe annualized alphas remain close to the corresponding net returns. Modest dollar-volume and capitalization restrictions, however, make both returns and alphas negative. The ranking is not simply a repackaging of conventional factors, but most of its measured economic value comes from stocks for which omitted costs and capacity constraints are likely most severe.

The concentration in small and illiquid stocks resembles the established economics of weekly reversal (Chen et al. 2025). Without an exact common-panel score comparison, however, we cannot conclude that the learned models reproduce the reversal signal.

4.3. Recent-sample fragility

Table 6 compares weekly performance across the baseline and extension periods, with model parameters, benchmark definitions, and portfolio rules held fixed. The baseline covers 2001–2019, while the extension covers 2020–2024. Because the extension spans only five years, its estimates should be read as a stability diagnostic.

Table 6: Weekly equal-weighted performance in the baseline and extension periods

	2001–2019				2020–2024			
Method	Gross ret.	Gross SR	Net ret.	Net SR	Gross ret.	Gross SR	Net ret.	Net SR
Chart CNN $I20/R5$	96.3%	7.64	60.0%	4.78	50.7%	3.87	18.5%	1.35
CNN1D $I20/R5$	102.2%	8.09	66.5%	5.28	49.9%	3.34	18.6%	1.18
Multimodal $I20/R5$	97.6%	7.76	64.4%	5.12	49.5%	3.19	17.2%	1.11
MOM	−5.1%	−0.19	−10.0%	−0.36	8.7%	0.26	4.1%	0.12
STR	48.7%	2.04	31.6%	1.33	27.7%	1.10	11.0%	0.44
WSTR	66.2%	3.33	31.7%	1.59	32.9%	1.42	−0.8%	−0.03
TREND	64.4%	3.09	38.3%	1.85	8.1%	0.38	−13.3%	−0.62

Note. Returns and Sharpe ratios are annualized. The learned rows use the representative 20-day input. Benchmark definitions appear in Section 2.4. Portfolio and cost conventions are fixed across periods.

All three learned portfolios weaken sharply after 2019. Their net Sharpe ratios fall from roughly five in the baseline sample to 1.11–1.35 in the extension, although net annualized returns remain positive near 18%. WSTR and TREND fall from net Sharpe ratios above 1.5 to zero or below, and STR also weakens. The post-2019 decline is therefore not unique to one architecture; it also appears in transparent short-horizon price rules.

Chart-CNN performance reaches its low point in 2020 and partially recovers during 2021–2024, as shown in the appendix. The five-year average consequently combines a sharp initial break with uneven later performance. This path resembles the recent attenuation reported by Kaczmarek and Pukthuanthong (2025) and broader predictor decay (McLean and Pontiff 2016), but we cannot distinguish market change from sampling variation or specification search.

The positive equal-weighted extension returns coexist with negative value-weighted net Sharpe ratios and full-sample losses under modest tradability screens. Some return-ranking ability remains after 2019, but at a smaller magnitude and with a weaker practical interpretation.

5. Conclusion

We ask which parts of chart construction produce the return ranking and where that ranking retains economic value. Building on the controlled comparison in Jiang et al. (2023), we compare representations designed to assess the roles of local high–low scaling, binary image layout, and temporal learning. We then compare scores directly, combine representations, restrict the tradable universe, and examine recent performance.

Representation matters before a flexible model is introduced. Local high–low scaling makes a strong weekly ranking accessible even to the logistic classifier, and CNN1D applied to the same continuous, locally scaled sequence closely matches the chart CNN in the strongest weekly specifications. Temporal convolution contributes most at the weekly horizon. This reinforces Jiang et al.'s finding that locally scaled sequences can reproduce much of the chart result and shows more fully that the binary image is not necessary to recover its weekly portfolio performance.

Chart CNN, CNN1D, and multimodal scores rank many of the same stocks, especially in the strongest weekly specifications, but their rankings are not identical and each score retains conditional variation in parts of the grid. Neither feature-level nor prediction-level fusion consistently improves on the strongest individual branch. The representations therefore overlap substantially but incompletely and offer little portfolio-level complementarity.

Learned portfolios outperform transparent price rules in the unrestricted equal-weighted universe and remain positive under the common transaction-cost schedule. Net performance nevertheless turns negative under value weighting and after modest dollar-volume or market-capitalization screens. These screens are not direct counterparts to Jiang et al.'s largest-500 exercise. Conventional factor adjustment does not absorb the unrestricted spread, but the returns are concentrated where market impact, borrowing constraints, and capacity are most concerning. All learned models also weaken after 2019, alongside short-horizon price-based benchmarks, although the five-year extension is too short to identify the source of the decline.

Most of the chart CNN's historically strong weekly portfolio performance can therefore be recovered from locally scaled continuous sequences with temporal convolution, without the binary image layout. Its measured economic value is concentrated in difficult-to-trade stocks and weakens after 2019. Chart and sequence scores are not identical, but the binary chart adds little portfolio performance once the same local scaling and temporal structure are available.

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